MILITARY SPECIFICATION

GAGE, COMPOUND PRESSURE-VACUUM,

DIAL INDICATING

This limited coordination Military specification has been prepared by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa. 19101, based upon currently available technical information, but it has not been approved for promulgation as a coordinated document. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used for procurement.

1. SCOPE

1.1 This specification covers one type of elastic chamber, circular face, dial-indicating gage suitable for determining the pressures both above and below atmospheric inside of sealed cans of food.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water-Resistant (for Sealing Fiberboard Boxes).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (for Carton Sealing).

Military

MIL-P-116	Preservation, Methods of.

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STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

10167	Gage, Compound Pressure-Vacuum, Dial Indicating.

(Copies of Military specifications, standards and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN IRON AND STEEL INSTITUTE

Stainless Steel

(Application for copies should be addressed to American Iron and Steel Institute, 150 East 42nd Street, New York, New York 10017.)

AMERICAN NATIONAL STANDARDS INSTITUTE (A.N.S.I.)

A.N.S.I.	Gages - Pressure and Vacuum, Indicating Dial Type - Elastic Element.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Designation D 2240-68	Standard Method of Test for: Indentation Hardness of Rubber and Plastics by Means of a Durometer.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

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3. MATERIAL

3.1 Case, bezel ring, knurl nut and external thread connection. The case, bezel ring, knurl nut and external threaded connection shall be fabricated from nickel-plated brass or AISI 300 series stainless steel. The plating and finish shall be uniform and free from scratches, chipping, peeling, and discoloration.

3.2 Piercing device. The piercing device shall be fabricated from a 300 series stainless steel hollow tube or AISI 1090 steel.

3.3 Grommet. The grommet shall be fabricated from rubber. The rubber shall have a durometer hardness of A/35-40/1 when tested as specified in 4.4.2.

3.4 Dial crystal. The dial crystal shall be fabricated from glass.

3.5 Style and design. The "Bourdon-Tube Elastic Chamber" shall be housed in a case to which the dial crystal shall be secured by a threaded bezel ring. The stem shall have a 1/8-27 ANPT male pipe thread which shall be fitted to the steel can piercing device. The piercing device shall be 1/2 ± 1/16 inch long and shall be cut at an angle of 30° to the body. A grommet shall be designed to fit over the point of the piercing device which shall provide an airtight seal while the can is being punctured. The knurl nut shall be screwed on to the base of the stem and shall act as a hub which is to prevent the grommet from being pushed out of position.

3.6 Dial face. The dial face shall be designed to indicate Vacuum (VAC) and Pressure (PRESS) and shall be in accordance with Defense Personnel Support Center, DPSC Drawing No. 10167. The numbers and graduation marks shall be black on a silver colored background. Pointer shall be securely attached and shall be free from bends.

3.7 Accuracy . The accuracy of the gage shall be determined as specified in 4.4.1.

3.8 Identification marking. The dial face shall be legibly and permanently marked with the manufacturer's name or registered trademark.

3.9 Workmanship. Workmanship shall be first class throughout. The finished gages shall be free from defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the rubber and metal, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

Table I. Sampling for Examination

	Inspection Level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-1	6.5

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4.2.2 For tests. Sampling for accuracy test shall be conducted in accordance with MIL-STD-105 for inspection level S-2. Acceptance number shall be zero.

4.3 Examination. The gages shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be in accordance with the following classification of defects:

Categories and defects*

Major		
101		Gages not free from dents, jagged edges and burrs.
102		Cutting edge of piercing device not smoothly finished.
103		Dial pointer not securely attached to the shaft and not free from bends.
104		Component parts missing, damaged, or inoperable.
105		Piercing device not capable of piercing can.
106		Glass crystal not free of chips, cracks, and scratches.
107		Plating not of uniform thickness and not free from scratches, peeling or discoloration.
Minor		
201		Identification markings missing or not legible.
202		Gages not free from grease or dirt.

4.3.2 Dimensional examination. The gages shall be examined for defects in dimensions. Any dimension not within tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accuracy. The gages shall be tested for accuracy in accordance with A.N.S.I. B40.1 for "Grade B Gages."

4.4.2 Durometer hardness. The durometer hardness of the rubber grommet shall be determined in accordance with ASTM D 2240-68.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One gage, as specified, constitutes one unit of issue.

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5.1.2 Packaging quantities. The number of units of issue indicated in the following table shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table II. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	6 units	72 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Not required.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive materials.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IA-15.

5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Gage is intended for use in determining the vacuum or pressure inside of sealed food cans.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
(c) Federal stock number (FSN).

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6.3 This specification is applicable to the following item listed in the Federal Supply Catalog:

6685-806-9300	Gage, Compound Pressure-Vacuum, Dial Indicating.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6685-M389

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R755

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements .

SPECIFICATION
MIL-G-36721(DSA-DM) FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. WAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING.

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID .

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional) | DATE

DD FORM 1 JAN 88 1426 REPLACES EDITION OF 1 OCT 84, WHICH MAY BE USED.

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